# Electromagnetic induction

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain why changing magnetic flux can induce an emf.

**SUCCESS CRITERIA - I can...**

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A 3 Ω resistor carries 5 A. Find power using both IV and $I^2R$, showing the implied voltage.

**Marzano check - circle:** ① ② ③

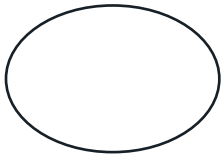
## 02 Worked example: explain the model

**Reference:** $\Phi = BA$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

**Worked reasoning:** A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

**Your turn:** Draw a magnet approaching a coil and mark the changing field through the loop. Contrast it with a stationary case.

Label field direction, loop area and what changes.



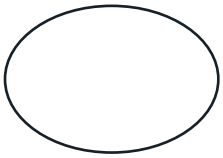
Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

Compare a magnet held still beside a fixed coil with a magnet moving into the coil. Which can produce a changing flux?

Label field direction, loop area and what changes.



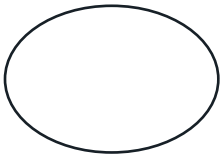
Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A 100-turn lab coil has area 0.020 $m^2$. A uniform field perpendicular to it changes from 0.10 T to 0.30 T in 0.50 s. Label the flux model and calculate average induced-emf magnitude.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** ① ② ③

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Continue down each column.

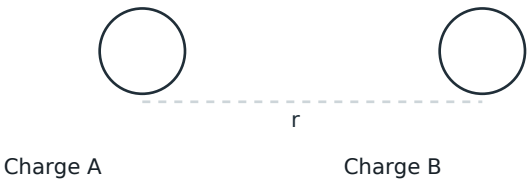
## 05 Explain your reasoning

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

Both charge magnitudes become 4 times as large and separation becomes 6 times as large. Express new force as a fraction of the old.
Label charges and both interaction forces.

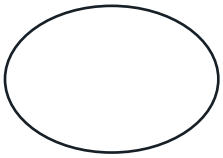



**Marzano check - circle:** 1 2 3

## 07 Exit check

The same flux change occurs in one fourth the time. Compare emf magnitude and name a model assumption.

Label field direction, loop area and what changes.



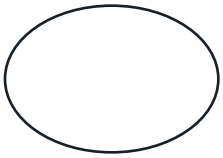
Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

Flux per turn changes from 0.01 to 0.04 Wb in 0.2 s through 50 turns. Find average emf magnitude and describe Lenz's-law direction qualitatively.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

**Marzano check - circle:** 1 2 3